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%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%禁忌搜索算法解决TSP问题%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%初始化%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
clear all; %清除所有变量
close all; %清图
clc; %清屏

C = [1304 2312;3639 1315;4177 2244;3712 1399;3488 1535;3326 1556;
     3238 1229;4196 1044;4312 790;4386 570;3007 1970;2562 1756;
     2788 1491;2381 1676;1332 695;3715 1678;3918 2179;4061 2370;
     3780 2212;3676 2578;4029 2838;4263 2931;3429 1908;3507 2376;
     3394 2643;3439 3201;2935 3240;3140 3550;2545 2357;2778 2826;
     2370 2975]; %31个省会城市坐标
N=size(C,1); %TSP问题的规模,即城市数目
D=zeros(N); %任意两个城市距离矩阵
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%求任意两个城市距离矩阵%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
for i=1:N
    for j=1:N
        D(i,j)=((C(i,1)-C(j,1))^2+(C(i,2)-C(j,2))^2)^0.5;
    end
end
Tabu=zeros(N); %禁忌表
TabuL=round((N*(N-1)/2)^0.5); %禁忌长度
Ca=200; %候选邻域解集大小(全部邻域解个数)
CaNum=zeros(Ca,N); %候选邻域解集合
S0=randperm(N); %随机产生初始解
bestsofar=S0; %当前最佳解
BestL=Inf; %当前最佳解距离
figure(1);
p=1;
Gmax=1000; %最大迭代次数
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%禁忌搜索循环%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
while p<Gmax
    ALong(p)=func1(D,S0); %当前解适值
    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%交换城市%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
    i=1;
    A=zeros(Ca,2); %解中交换的城市矩阵
    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%求领域解中交换的城市矩阵%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
    while i<=Ca
        M=N*rand(1,2);
        M=ceil(M);
        if M(1)~=M(2)
            A(i,1)=max(M(1),M(2));
            A(i,2)=min(M(1),M(2));
            if i==1
                isa=0;
            end
        end
        i=i+1;
    end
    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%更新最佳解%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
    if ALong(p)<BestL
        BestL=ALong(p);
        S0=S0(A(p,1):A(p,2));
        bestsofar=S0;
    end
    p=p+1;
end

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else
    for j=1:i-1
        if A(i,1)==A(j,1) && A(i,2)==A(j,2)
            isa=1;
            break;
        else
            isa=0;
        end
    end
end
if ~isa
    i=i+1;
else
end
else
end
end
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%产生领域解%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%保留前BestCaNum个最好候选解%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
BestCaNum=Ca/2;
BestCa=Inf*ones(BestCaNum,4);
F=zeros(1,Ca);
for i=1:Ca
    CaNum(i,:)=S0;
    CaNum(i,[A(i,2),A(i,1)])=S0([A(i,1),A(i,2)]);
    F(i)=func1(D,CaNum(i,:));
    if i<=BestCaNum
        BestCa(i,2)=F(i);
        BestCa(i,1)=i;
        BestCa(i,3)=S0(A(i,1));
        BestCa(i,4)=S0(A(i,2));
    else
        for j=1:BestCaNum
            if F(i)<BestCa(j,2)
                BestCa(j,2)=F(i);
                BestCa(j,1)=i;
                BestCa(j,3)=S0(A(i,1));
                BestCa(j,4)=S0(A(i,2));
                break;
            end
        end
    end
end
end
[JL,Index]=sort(BestCa(:,2));
SBest=BestCa(Index,:);

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BestCa=SBest;
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%破禁准则%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
if BestCa(1,2)<BestL
    BestL=BestCa(1,2);
    S0=CaNum(BestCa(1,1),:);
    bestsofar=S0;
    for m=1:N
        for n=1:N
            if Tabu(m,n)~=0
                Tabu(m,n)=Tabu(m,n)-1;
                %更新禁忌表
            end
        end
    end
    Tabu(BestCa(1,3),BestCa(1,4))=TabuL;
    %更新禁忌表
else
    for i=1:BestCaNum
        if Tabu(BestCa(i,3),BestCa(i,4))==0
            S0=CaNum(BestCa(i,1),:);
            for m=1:N
                for n=1:N
                    if Tabu(m,n)~=0
                        Tabu(m,n)=Tabu(m,n)-1;
                        %更新禁忌表
                    end
                end
            end
            Tabu(BestCa(i,3),BestCa(i,4))=TabuL;
            %更新禁忌表
            break;
        end
    end
end
ArrBestL(p)=BestL;
p=p+1;
for i=1:N-1
    plot([C(bestsofar(i),1),C(bestsofar(i+1),1)],...
        [C(bestsofar(i),2),C(bestsofar(i+1),2)],'bo-');
    hold on;
end
plot([C(bestsofar(N),1),C(bestsofar(1),1)],...
    [C(bestsofar(N),2),C(bestsofar(1),2)],'ro-');
title(['优化最短距离:',num2str(BestL)]);
hold off;

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        pause(0.005);
    end
    BestShortcut=bestsofar;           %最佳路线
    theMinDistance=BestL;            %最佳路线长度
    figure(2);
    plot(ArrBestL);
    xlabel('迭代次数')
    ylabel('目标函数值')
    title('适值迭代曲线')

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%适值函数%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
function F=func1(D,s)
    DistanV=0;
    n=size(s,2);
    for i=1:(n-1)
        DistanV=DistanV+D(s(i),s(i+1));
    end
    DistanV=DistanV+D(s(n),s(1));
    F=DistanV;#

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